

THANG NGUYEN HUU

Senior FEA Engineer — Nonlinear Finite Element Analysis & Structural Performance Evaluation

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PROFESSIONAL SUMMARY

Structural Analysis Engineer specializing in nonlinear finite element analysis (FEA) and structural performance evaluation. Experienced in advanced modeling of contact behavior, material nonlinearity, and stability analysis with strong correlation to experimental testing. Published author with a solid computational mechanics foundation and industry experience supporting product qualification and design optimization.

TECHNICAL SKILLS

Finite Element Analysis

Abaqus ANSYS Nonlinear Analysis Contact Modeling Material Nonlinearity
Stability / Buckling Heat Transfer Test Correlation

Programming

C# / .NET / WPF / MVVM Python (Abaqus Scripting) C++ / Qt VBA MATLAB

Engineering Tools

AutoCAD Siemens NX SolidWorks Inventor CAD Automation LaTeX Documentation

PROFESSIONAL EXPERIENCE

Senior FEA Engineer

01/2025 — Present

Simpson Strong-Tie · Ho Chi Minh City

- Lead nonlinear finite element simulations for structural connectors and anchoring systems, ensuring strong correlation with full-scale physical testing.
- Review and validate complex FEA models to ensure technical accuracy and compliance with internal and external standards.
- Coordinate simulation planning and task allocation across cross-functional teams to meet project deadlines.

- Develop internal automation tools and engineering applications to improve modeling efficiency and workflow reliability.
- Provide technical mentorship and training on advanced FEA methodologies and best practices.

○ FEA Engineer

05/2021 — 12/2024

Simpson Strong-Tie · Ho Chi Minh City

- Performed nonlinear FEA simulations in Abaqus for structural connectors and anchoring systems under static and stability loading conditions.
- Developed Python-based Abaqus plug-ins to automate model generation and post-processing.
- Correlated numerical simulations with experimental data to validate structural performance.
- Prepared technical reports and supported internal knowledge-sharing initiatives.

SELECTED PROJECTS

Heat Transfer Parameter Identification Using Inverse Analysis

- Custom 2D heat transfer finite element solver in MATLAB.
- Grey Wolf Optimization for inverse parameter identification.
- POD-RBF model reduction cutting computational cost significantly.

MATLAB

FEM

Optimization

POD-RBF

FEA Viewer (C++ / Qt)

- MVC desktop app for reading and visualizing Abaqus *.odb files.
- C++ data conversion kernel using the Abaqus API.
- VTK-based 3D interactive visualization and post-processing.

C++

Qt

Abaqus API

VTK

WinUX (C# / WPF)

- MVVM desktop app for secure file transfer (SFTP) between Windows and Linux servers.
- Remote job monitoring and execution control for simulation workflows.
- Real-time communication features improving server interaction efficiency.

C#

WPF

MVVM

SFTP

Engineering Automation

- Python Abaqus plug-ins for automated model generation and post-processing.
- CAD automation and workflow tools improving modeling reliability.

Python

Abaqus Scripting

CAD Automation

OPEN SOURCE

Selected public repositories — engineering tools, numerical libraries, and FEA utilities.

AnsysReader

C++17

Native readers and writers for ANSYS result and database files — structural .rst parsing (mesh, results, cyclic data), EMAT / FULL / CDB / DAT support, v1.0.0 public API baseline.

FEAViz

C17

Modular visualization toolkit for scientific, engineering, and CAE applications — OpenGL rendering, FEA/CFD modules, VTK / STL / OBJ I/O, Python bindings and Qt adapters.

XLPP

C++20

High-performance Excel library — reads/writes .xlsx, .xls, .xlsb, .csv with SIMD XML parsing and multi-threaded save; MIT licensed with Python / C# / C bindings (PyPI, NuGet).

Numericals

C++14

Lightweight NumPy-inspired numerical library — NdArray, Vector, and Matrix API with elementwise ufuncs and linear-algebra helpers.

PdfPP

C++

C++ PDF library (Pdf++) with cross-engine validation suite, fuzzing corpus, and a Windows viewer application.

[View all repositories on GitHub →](#)

PUBLICATIONS

Nguyen, M. N., **Nguyen, T. H.**, & Bui, T. Q. (2023). Finite element modeling of tapered concrete-filled steel tubular columns under axial compression. *VNUHCM Journal of Engineering and Technology*, 4(4). [doi:10.32508/stdjet.v4i4.937](https://doi.org/10.32508/stdjet.v4i4.937)

EDUCATION

B.Eng. in Engineering Mechanics

2021

Ho Chi Minh City University of Technology, Vietnam

Relevant coursework: Finite Element Method, Structural Analysis, Numerical Methods.

LANGUAGES

English — Professional Working Proficiency · Vietnamese — Native